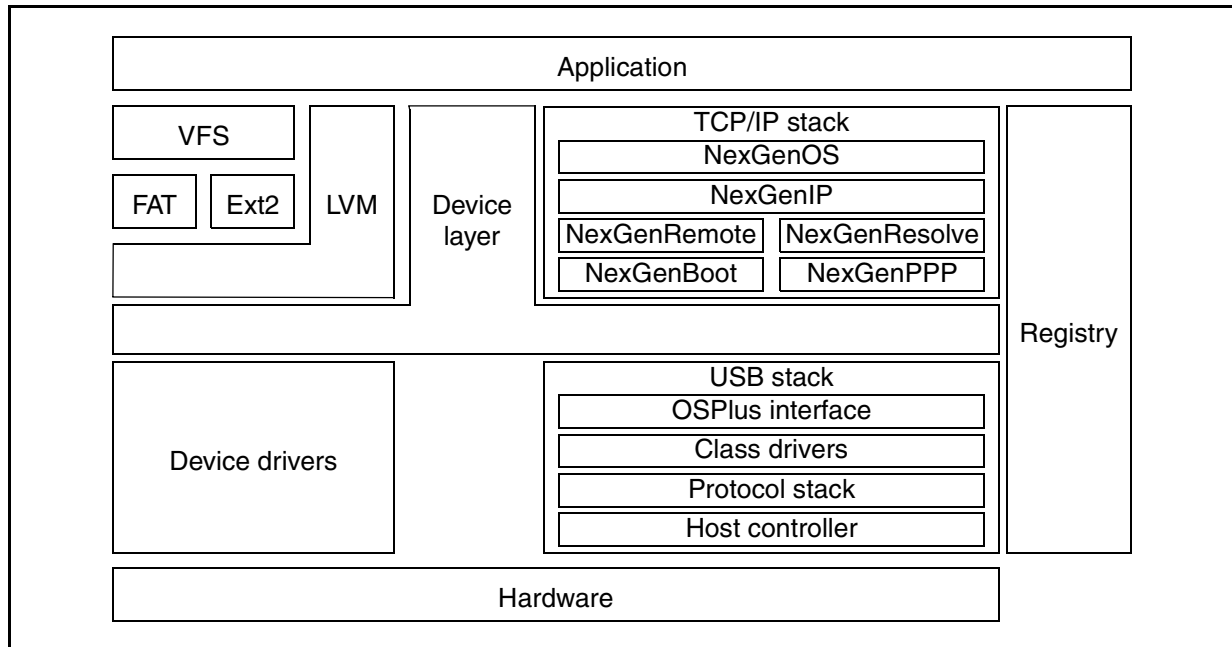


Enhancement package for OS20/OS21

Data brief



Description

OSPlus is an enhancement package for the OS20 and OS21 operating systems. It provides a number of new APIs that enable a device driver and I/O infrastructure. OSPlus is delivered as a base package and provides support for the OSPlus TCP/IP and OSPlus USB extension packages.

The OSPlus base package extends OS20 (on ST20) and OS21 (on ARM, ST40 and ST200) to include a device driver infrastructure, various device drivers, a file system infrastructure including a virtual file system (VFS) API, and implementations of FAT and EXT2 file systems.

OSPlus extensions provide system software extensions such as a TCP/IP stack and a USB stack. These extension packages require a suitable OSPlus base package to be installed as a prerequisite as they make use of the registry and other aspects of the OSPlus infrastructure.

Features

- Standard device driver API
- Device driver writer's interface
- Registry services
- Drivers for standard devices, including:
 - PCI
 - serial ports
 - GPDMA
 - SMSC LAN91C111 and LAN9x1x Ethernet
 - SATA and ATA hard disks
- Virtual file system (VFS) API through which multiple file systems can be accessed
- E2FS: Ext2 file system (Linux-style file system)
- FAT12/16/32 (PC-style file system)
- Logical volume manager supporting hard disks and removable media
- Supports OSPlus TCP/IP extension package
- Supports OSPlus USB extension package

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1 Introduction

OSPlus is an enhancement package for the OS20 and OS21 operating systems comprising the OSPlus base package and extensions. It includes new APIs that enable a device driver infrastructure.

This datasheet describes the main components of the OSPlus packages.

1.1 OSPlus base package

An OSPlus base package contains the binaries for the ARM and for STMicroelectronics ST20, ST40 and ST200 CPU types, plus documentation. The ST20 support includes binaries for both ST20-C2 and ST20-C1. The ST200 support includes binaries compiled for the ST220 and the ST231. Different variants of the package are available for different host types (Microsoft Windows, Red Hat Linux and Solaris). There are no technical differences between these; only the format of text files and the form of the installation (Windows Zip file, Linux RPM file, or UNIX tar file).

The base package includes the following components:

- a standard device driver API
- a device driver writer's interface
- registry services
- drivers for standard devices, including:
 - PCI
 - serial ports
 - GPDMA
 - SMSC LAN91C111 and LAN9x1x Ethernet
 - SATA and ATA hard disks
- single API virtual file system (VFS) through which multiple file systems can be accessed
- E2FS, Ext2 file system (Linux-style file system)
- FAT12/16/32 (PC-style file system)
- logical volume manager supporting hard disks and removable media

1.2 OSPlus extension packages

The extension packages require a suitable OSPlus base package to be installed as a prerequisite. The extensions make use of the registry and other aspects of the OSPlus infrastructure.

1.3 OSPlus TCP/IP extension package

The OSPlus TCP/IP extension package is based on the NexGen TCP/IP stack architecture. Separate products are available for ST20, ST40 and ST200.

Note: The OSPlus TCP/IP package is not available for the ARM.

The TCP/IP package contains the following products:

- NexGenOS: OS abstraction layer
- NexGenIP: TCP/IP stack
- NexGenBOOT: BootP, TFTP, DHCP
- NexGenPPP: PPP and PPPoE (PPP over Ethernet)
- NexGenREMOTE: FTP client and server, Telnet

The NexGenRESOLV product (DNS resolver) previously included in the ST20 TCP/IP package is now incorporated into the NexGenIP product.

The libraries are provided in binary form. Each product is delivered on CD-ROM that includes installation files for all three supported host operating systems, plus the installation files for the required base OSPlus packages.

The OSPlus base package provides support for:

- PPP and SLIP through the onboard serial ports
- Ethernet access through SMSC LAN91C111, SMSC LAN9x1x and STMAC110

The OSPlus USB package provides support for:

- PPP and SLIP through USB serial adapters
- Ethernet access through USB Ethernet and host-to-host adapters

For more detailed information about the TCP/IP OSPlus extension package, refer to the *OSPlus TCP/IP Extension data brief* (ADCS 7816153).

1.4 OSPlus USB extension package

The OSPlus USB extension package provides support for the integrated USB 1.1 and 2.0 host controllers on ST devices. This allows devices to be hot plugged to and from the USB.

The USB extension package supports the following built-in device classes:

- hub
- human interface devices (HID) for example, mouse, keyboard and joystick
- mass storage
- communications (Ethernet adapters, cable modem, serial)
- host-to-host

An interface is provided to enable the writing of new class drivers for devices that are not directly supported.

The USB stack libraries are provided in binary form. Each product is delivered on a CD-ROM that includes installation files for all three supported host operating systems, plus the installation files for the required base OSPlus packages.

For more detailed information about the USB OSPlus extension package, refer to the *OSPlus USB 2.0 Extension data brief* (ADCS 7801961).

1.5 OSPlus toolset version requirements

The OSPlus base and extension packages can be used with the toolsets shown in [Table 1](#).

Table 1. Toolset requirements

CPU type	Toolset
ARM	Use with ARM RVDS toolset 3.1-patch 1 or later
ST20	Use with ST20 toolset R2.0 or later
ST40	Use with ST40 toolset R3.0 or later
ST200	Use with ST200 toolset R4.1 or later

2 Ordering information

The OSPlus base package is available for download free of charge. For ordering information, contact your nearest STMicroelectronics regional sales office or local FAE. For the OSPlus extension packages, the ordering information is as follows:

Table 2. Ordering Information

OSPlus extension product	Product codes
OSPlus ST20 TCP/IP	ST20-TCPIP
OSPlus ST40 TCP/IP	ST40-TCPIP
OSPlus ST20 USB	STORM-USB20/20
OSPlus ST40 USB	STORM-USB20/40

Acknowledgements

The OSPlus TCP/IP extension is based on technology developed by NexGen.
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Revision history

Table 3. Document revision history

Date	Revision	Changes
15-Feb-2011	G	Minor changes to formatting
11-Nov-2009	F	Minor changes to formatting Changes to ARM requirements in Table 1 .
5-Nov-2007	E	Updated to new template
21-Sep-2006	D	Throughout: SATA and ATA drivers added and 1394 stack information removed OSPlus TCP/IP extension page: Added details of USB extension package
29-Sep-2005	C	Table 1: For CPU type ST200, changed toolsets from R4.0 or later to R4.1 or later Introduction: Removed 9118 support as a future feature as it's now included as standard
23-Sep-2005	B	Table 1: For CPU type ST40, changed toolsets from R2.1 or later to R3.0 or later
11-May-2005	A	Initial release.

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